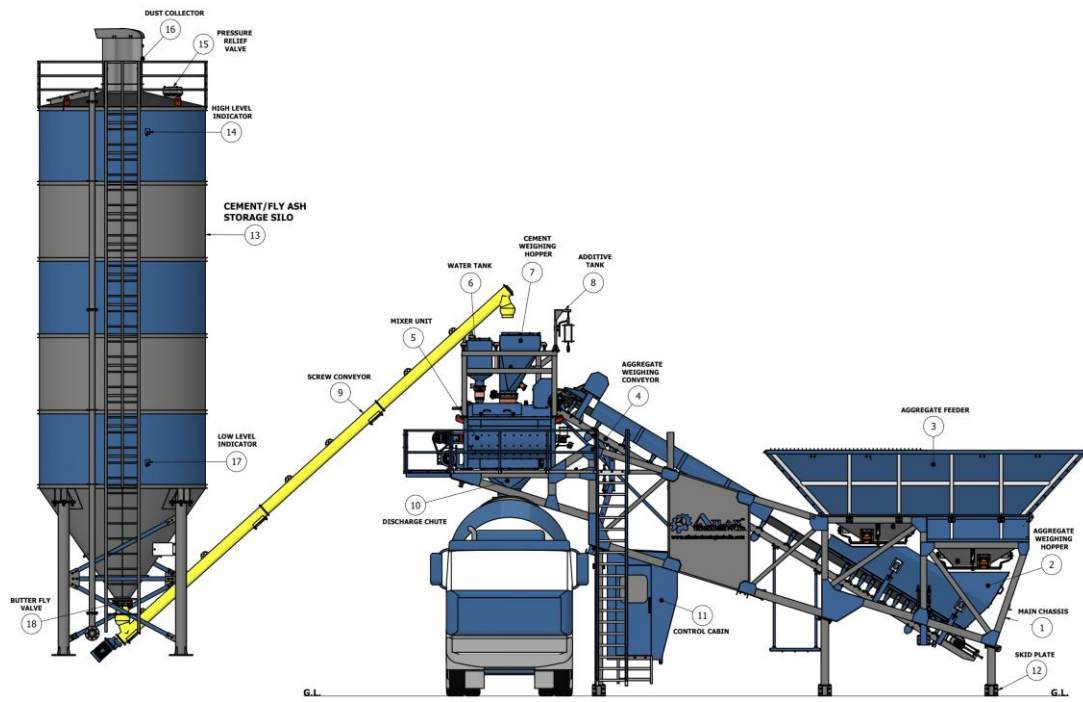




### DETAIL FORM FOR MOBILE CONCRETE BATCHING PLANT WITH TWIN SHAFT MIXER

**BELOW REFERENCE IS 75 M<sup>3</sup> PER HOUR**

- |        |                                                                                              |
|--------|----------------------------------------------------------------------------------------------|
| Item 1 | four (4) bin feeder with individual bin , conveyors, weighing/mixing section & control cabin |
| Item 2 | 100 tons cement silo with accessories ( Optional )                                           |
| Item 3 | cement feeding system for cement silo( Optional )                                            |
| Item 4 | Plant control system                                                                         |
| Item 5 | Wiring                                                                                       |
| Item 6 | Supervisory installation and commissioning                                                   |

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## TECHNICAL SPECIFICATION

### PLANT CAPACITY

**For Reference: 75 m<sup>3</sup> based on a mixer capacity of 1875 / 1250 litter. Of vibrated concrete per batch 1.25 m<sup>3</sup> and at (60) second weigh/mix cycle.**

### Assuming the following conditions: -

- 1) 100% Plant utilisation
- 2) Material density ( 2400 ) Kg/m<sup>3</sup>

### **1 INDIVIDUAL 2/ 4 BIN FEEDERS, CHARGING (WEIGH) CONVEYORS & CABIN**

#### **1.1 AGGREGATE HOPPERS**

|                 |   |                                                                            |
|-----------------|---|----------------------------------------------------------------------------|
| Feeders         | - | Four (4) individual Bin                                                    |
| Capacity        | - | 10 m <sup>3</sup> /each bin                                                |
| Loading width   | - | 2.7 meter                                                                  |
| Loading height  | - | 6 meter                                                                    |
| Material        | - | 5 mm thick mild steel plate                                                |
| Vibrator        | - | One (1)                                                                    |
| Discharge gates | - | Four (4) pneumatically controlled discharge gates,<br>One gates per hopper |

#### **1.2 CHARGING (WEIGH) CONVEYOR**

|                        |   |                                                                  |
|------------------------|---|------------------------------------------------------------------|
| Weigh conveyor         | - | Mounted under weigh hoppers                                      |
| Length                 | - | 8.35 Metres                                                      |
| Belt                   | - | 800 mm wide 4 ply with vulcanised joint                          |
| Idlers                 | - | Troughed bolted to steel section support frame                   |
| Drive                  | - | 20 HP gear motor unit direct on head shaft                       |
| Belt tensioning        | - | Tension bolts fitted to tail drum slide bearings                 |
| Belt scraper           | - | Fitted under head drum                                           |
| Weigh hopper           | - | Manufactured from 5 mm mild steel plate                          |
| Weighing capacity      | - | 1.5 m <sup>3</sup> total from all hoppers, mounted on load cells |
| Load cells             | - | 2000 kg. "S" type - 4 no.                                        |
| Transfer load capacity | - | 150 Kg. cm <sup>2</sup>                                          |

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**1.4**

**CEMENT WEIGH HOPPERS**

|                 |   |                                                                         |
|-----------------|---|-------------------------------------------------------------------------|
| Capacity        | - | One 850 Kg. load cell mounted                                           |
| Load cells      | - | 500 kg. Shear beam type - 3 no.                                         |
| Plate thickness | - | 3 & 5 mm steel plate                                                    |
| Discharge doors | - | 300 mm pneumatically operated butterfly valves with On-Off limit switch |
| Vibrator        | - | 0.33 HP vibrating motor                                                 |
| Total inlets    | - | Two for optional customer supply flyesh silo / feed screw               |

**1.5**

**WATER WEIGH HOPPER**

|                 |   |                                                                  |
|-----------------|---|------------------------------------------------------------------|
| Capacity        | - | 400 liter. Load cell mounted                                     |
| Load cells      | - | 300 kg. "S" type - 2 no.                                         |
| Plate thickness | - | 3 mm mild steel plate                                            |
| Discharge valve | - | 100 mm pneumatically operated butterfly valve with On-Off switch |

**1.6**

**ADDITIVE WEIGH HOPPER**

|                 |   |                                               |
|-----------------|---|-----------------------------------------------|
| Capacity        | - | 10 liter x 2 load cell mounted                |
| Load cells      | - | 50 kg. "S" type - 1 no.                       |
| Material        | - | 5 mm thick and transparent acrylic dispenser. |
| Discharge valve | - | Spherical valve operated via actuator         |

**1.7**

**TWIN SHAFT MIXER (1875/1250) – ATLAS MAKE (SICOMA OPTIONAL)**

- VIBRATED MATERIAL YIELD 1250 LITRES PER CYCLE
- WITH AUTOMATIC DISCHARGE DOOR, HYDRAULIC OPENING
- WITH EMERGENCY MANUAL PUMP FOR DISCHARGE DOOR OPENING
- WITH MIXING BLADES IN WEAR RESISTING CAST IRON (600HB)
- WITH SCRAPER BLADES IN WEAR RESISTING CAST IRON (600HB)
- WITH BOTTOM LINERS IN WEAR RESISTING CAST IRON (16mm TH) (500HB)
- WITH LATERAL LINERS IN WEAR RESISTING CAST IRON (12mm TH) (500HB)
- HYDRAULIC POWER PACK MOTOR 1 x 1.5kW
- AUTOMATIC GREASE PUMP

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- DISCHARGE GATE THREE STAGES LIMIT SWITCHES
- INSPECTION GATE WITH SAFETY SWITCH
- JUNCTION BOXES FOR POWER SUPPLY AND CONTROL
- MIXING BLADE, 8 SETS
- SCRAPER BLADE, 4 SETS
- MIXING ARM, 12 SETS
- ARMS 90 DEGREE
- SANDBLASTING, UNDERCOAT PRIMER AND FINISH PAINT
- SINGLE OR DOUBLE ELECTRIC MOTOR 415 V, 3PH, 50HZ
- PLATFORM STEP

### 1.8 PNEUMATICS

- |            |   |                                                                     |
|------------|---|---------------------------------------------------------------------|
| Compressor | - | 5 HP to give 12 Kg. pressure with 17.32 CFM of piston displacement. |
| Pneumatics | - | Cylinder, Solenoid valves, nylon pipe work and fittings             |

### 1.9 CONTROL CABIN

- |              |   |                                                                                          |
|--------------|---|------------------------------------------------------------------------------------------|
| Type         | - | Tilted control cabin with hinge.                                                         |
| Construction | - | Corrugated sheet steel cladding and folds into under frame for transport                 |
| Base frame   | - | Constructed from rolled steel section                                                    |
| Roof         | - | Sheet steel. Roof void fully insulated with 30mm thermocol sheet.                        |
| Floor        | - | Wooden floor with carpet.                                                                |
| Walls        | - | Clad internally with decorative boarding. Walls fully insulated with 30 mm mineral wool. |
| Windows      | - | Three sides                                                                              |
| Lighting     | - | Overhead LED fitted.                                                                     |
| Heating      | - | Air conditioning unit (Optional)                                                         |
| Door         | - | Wide access door with door closer and lock                                               |
| Power        | - | Access from bottom of panel                                                              |

### 1.10 PLATFORMS

- |           |   |                                                                                                          |
|-----------|---|----------------------------------------------------------------------------------------------------------|
| Platforms | - | Maintenance platform for ease of access to mixer and weigh hoppers, with hand railing and access ladders |
|-----------|---|----------------------------------------------------------------------------------------------------------|

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**1.11**

**UNDER FRAME**

|                |   |                                              |
|----------------|---|----------------------------------------------|
| Under frame    | - | Fabricated rolled steel section chassis      |
| Support jacks  | - | Fixed supports under mixer                   |
| Loading height | - | 4.1 meter clearance under mixer outlet chute |

**1.12**

**2.5 TON CEMENT HOPPER**

Cement feeding hopper capacity 50 bags (2.5 T) with 219MM X 12000 MM screw conveyor.  
There are two support lag for screw conveyor mounting.

**2**

**100 T CEMENT SILO (OPTIONAL)**

- 2.1 100 Ton Capacity Silo for Cement** With fabricated 4 Column, Cross Bracings, Railing at Top, Climbing Ladder, Seamless Pipe (Dia: 100 mm Seamless pipe for feeding material in silo), Aeration System at Bottom of cone & with other necessary filaments.

**2.2**

**SILO ACCESSORIES (1 set)**

|                                                                                                          |   |           |
|----------------------------------------------------------------------------------------------------------|---|-----------|
| Double Flange Butterfly Valve with Cast Iron Disc Dia. 300 mm and Manual Lever for above Butterfly valve | - | One unit  |
| Special Dust Collector for Silo Venting                                                                  | - | One unit  |
| Level Indicator 115/230 V/AC Paddle type Rotating                                                        | - | Two units |
| Pressure Safety Relief Valve M.O.C. – M– M.S.                                                            | - | One unit  |
| Vibrating Bin aerator - (1 set inclusive of below items)                                                 |   |           |
| 1" Two way valve with coil                                                                               | - | 1 No.     |
| 1" Filter regulator with gauge                                                                           | - | 1 No.     |
| Bracket for FRL                                                                                          | - | 1 No.     |
| Manifold 1/2" female x 2 outlet.                                                                         | - | 1 No.     |
| Reducer 1" male X 1/2" female.                                                                           | - | 1 No.     |
| 1" Nipple-- 3 Nos                                                                                        |   |           |
| Straight connector 1/2" x 12mm OD                                                                        | - | 4 No.     |
| Equal tee 12mm OD                                                                                        | - | 2 No.     |
| PU tube 12mm OD                                                                                          | - | 10 Meter  |
| Male elbow 3/8" X 12mm OD                                                                                | - | 4 No.     |

Transfer screws - 219 mm diameter with 10 meters length cement screw conveyor



### 3 SILO FEEDING SYSTEM (OPTIONAL)

#### 3.1 Cement feeding compressor for bulker feeding

| SPECIFICATION   |                     |
|-----------------|---------------------|
| CAPACITY (FAD)  | 400(800 M3 / Hr)    |
| PRESSURE        | 1.0 Kg. / Cm2.      |
| MACHINE SPEED   | 1450 RPM            |
| POWER AT SHAFT  | 37 K.W.             |
| RECOMM. MOTOR   | 50 H.P. / 1450 RPM. |
| TYPE OF DRIVE   | V – BELT            |
| TYPE OF COOLING | AIR COOLED          |

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#### CONTROL SYSTEM

#### 4 CONTROLS

- 4.2 **CONTROL DESK** - Containing key switches for manual/auto control, manual Start/stop buttons and PLC weigh/mix control system
- 4.3 **WEIGH/MIX** - Weighing and mixing control system comprising of:  
Operator interface terminal with Sunlight Visible Colour TFT (5.7 INCH) Touch Screen Display (HMI - B&R OR DELTA) Super Bright 850 cd/M3 Display ensures maximum readability with a clear keyboard design and eight multifunction soft keys. Automatic control of the batching process via English interface with unlimited recipe storage. Printer supplied for production logging of essential data from each batch. Production data stored on non-volatile Compact Flash, accessible by USB cable. Network port to allow remote viewing and control of the batching process on customers PC.

#### OPTIONAL

**SCADA type control system with Wi-Fi internet attachment facility and also facility to operate plant by mobile / tablet.**

### 5 WIRING

Feeder and mixing unit section of mobile concrete plant is prewired with junction box. And from panel to genset is on customer's scope

#### **GUARDS**

Safety guards are provided over all V belt drives, chain drives and gears.

#### **STEELWORK**

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All welds to be cleaned as necessary, steelwork to be wire brushed and generally cleaned of all mill scale etc before painting.

#### **PAINTING**

All external surfaces are chipped and painted with one-coat primer  
With double coated PU paint.

#### **VOLTAGE**

400 Volt, 3 phase, 50 Hz

### **Mobile Concrete Plant Connected Power**

| Item | Quantity                           | HP   | Qty.         | Power (HP)    |
|------|------------------------------------|------|--------------|---------------|
| 1    | Twin shaft mixer                   | 30   | 2            | 60            |
| 2    | Weigh conveyor                     | 20   | 1            | 20            |
| 3    | Agg. hopper vibrator               | 0.5  | 1            | 0.5           |
| 4    | Cement weigh hopper vibrator       | 0.33 | 1            | 0.33          |
| 5    | Air compressor                     | 5    | 1            | 5             |
| 6    | Water pump                         | 3    | 1            | 3             |
| 7    | Additive motor                     | 1    | 1            | 1             |
| 8    | Hydraulic power pack               | 3    | 1            | 3             |
| 9    | Screw conveyor 219 mm x 14 meters  | 15   | 1            | 15            |
| 10   | Cement feeding system ( Optional ) | 50   | 1            | 50            |
|      |                                    |      | <b>TOTAL</b> | <b>160 HP</b> |

| Sr. No. | DESCRIPTION          | MAKE                                       |
|---------|----------------------|--------------------------------------------|
| 1       | Mixer gear box       | Brevini, Bonfiglioli, PBL, Top Gear        |
| 2       | All other gear boxes | Bonfiglioli/Wam/Prashant/Transolo/Top Gear |
| 3       | Screw conveyor       | WAM / IVA BULK                             |
| 4       | Motors               | HINDUSTAN/ Kirloskar / Crompton            |
| 5       | Conveyor belt        | MRF OR COMPERABLE                          |
| 6       | V belt               | Fenner                                     |
| 7       | Compressor           | Anest Iwata OR Indo Air                    |
| 8       | Pneumatics           | SMC                                        |
| 9       | Load cell            | SENSORTRONICS OR ASI OR KELLY              |
| 10      | Pedestal bearing     | TR                                         |
| 11      | Vibrating motors     | ISI Standard                               |
| 12      | Silo accessories     | IVA BULK / WAM                             |
| 13      | Water Pump           | Kirloskar or kebee                         |

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**Note: We recommend you to buy gen set of 300 kva. For the plant  
Main cable size: 70 sq. mm x 3 ½ core armoured copper – Length depending on the  
Generator distance.**

#### **Exclusions**

Civil works, foundations etc  
Mains power supply or generator (for usage of 250 KVA , 3 phases for grid and equivalent generator for power consumption on site )  
Mains incoming cable or distribution board  
Water supply and incoming pipe work  
Aggregate storage hopper grids  
Site labour and carnage  
Any item not specified

#### **Warranty**

The equipment is covered by guarantee under proper use, for faulty material or workman ship for a period of 12 months from the date of dispatch.

Consumable, electric components, rubber components and general wear and tear components are not covered under guarantee.

Purchaser shall inform supplier in writing, of any defect in equipment noticed during guarantee period. On receipt of purchaser's letter, supplier shall at his option, replace or repair free of cost the defective part supplied by him. In case repair is required to be done at works, purchaser shall send such defective part to the works after receiving supplier's confirmation.

We are in no event be liable for any damages whatsoever (including special, incidental consequential, direct or indirect damages, for personal injury, loss of business profits, or any other pecuniary loss) arising out of the use of or inability to use the equipment.